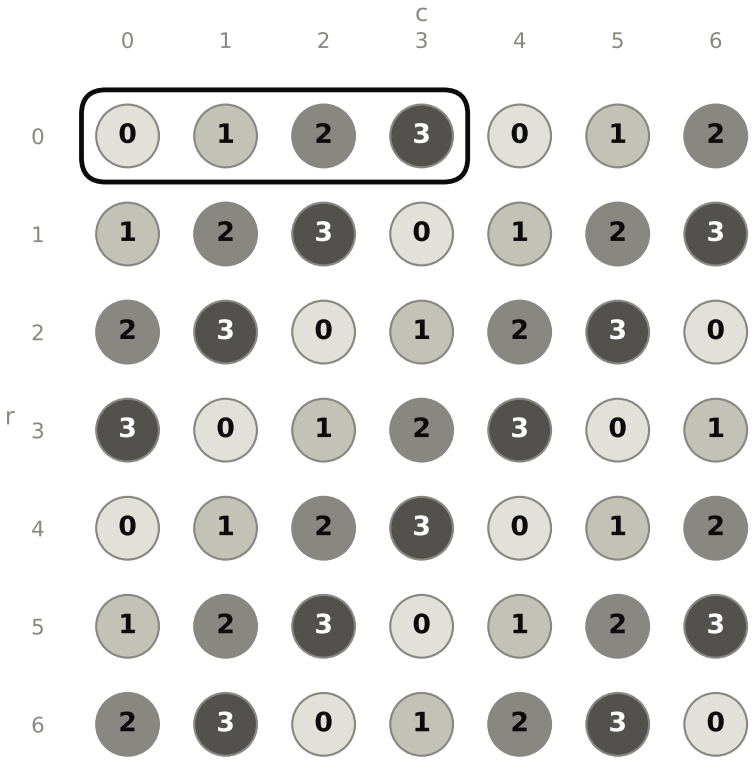
diag4: data qubit (c, r) takes letter k of the name's suffix (k = 0 is the first letter)

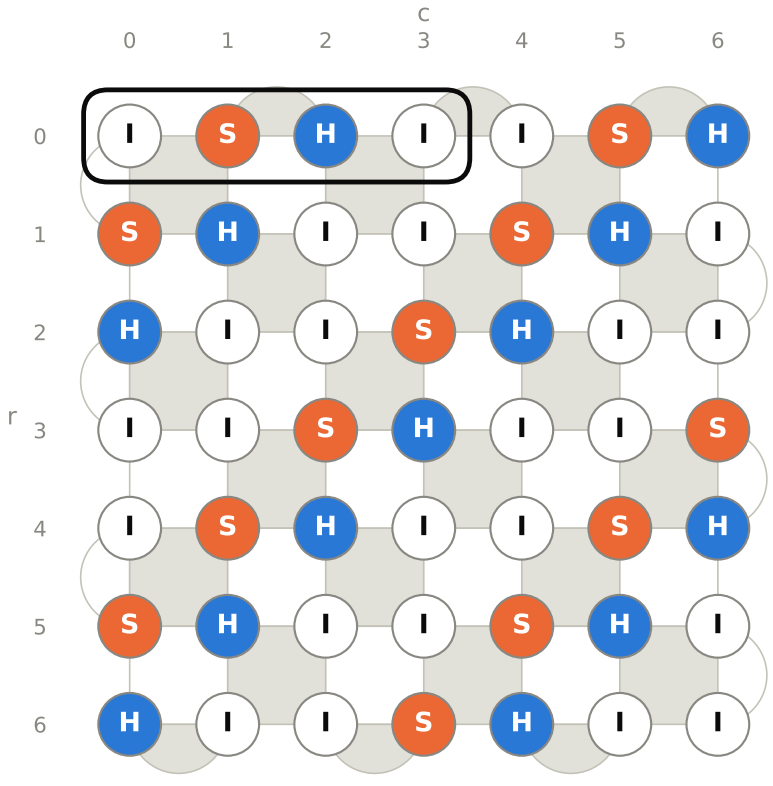
(c, r) = (0, 0) is the data qubit at (1, 1), top left; x = 2c + 1 grows to the right and y = 2r + 1 grows down, as in stim's diagrams

diag4: k = (r + c) mod 4



constant along every / diagonal

diag4_ISHI, d = 7



k = 0 → I, 1 → S, 2 → H, 3 → I

I: none; a Z error stays a Z error

H: Hadamard; a Z error acts as X on the CSS code

S: √X; a Z error acts as Y on the CSS code

Z check of the CSS code

X check of the CSS code

the qubits that spell the suffix in order

72 of 81 suffixes are registered; the others repeat an earlier code: IIII = rotated_surface, IHIH = xzzx, ISIS = cell_IS_SI, HIHI = cell_HI_IH, HHHH = cell_HH_HH, HSHS = cell_HS_SH, SISI = cell_SI_IS, SHSH = cell_SH_HS, SSSS = xy